Exelon Generation

10 CFR 50.73

LG-17-155
November 28, 2017

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Limerick Generating Station, Unit 2
Renewed Facility Operating License No. NPF-85
NRC Docket No. 50-353

Subject: LER 2017-007-00, Reactor Mode Change During RCIC Testing & Unit 2 LER Numbering

Enclosed is a Licensee Event Report (LER) which addresses a Mode Change with Reactor Core Isolation Cooling (RCIC) Testing in progress at Limerick Generating Station (LGS) Unit 2.

This LER is being submitted pursuant to the requirements of 10 CFR 50.73(a)(2)(i)(B), Operation or Condition Prohibited by Technical Specifications.

Additionally, this letter is informing the NRC per NUREG-1022, Revision 3, that LER numbers 001, 002, and 003 for docket 05000353 will not be used.

There are no commitments contained in this letter.

If you have any questions, please contact Robert B. Dickinson at (610) 718-3400.

Respectfully,

for Richard W. Libra
Vice President – Limerick Generating Station
Exelon Generation Company, LLC

cc: Administrator Region I, USNRC
USNRC Senior Resident Inspector, LGS

NRC FORM 366 (04-2017)	U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104	EXPIRES: 03/31/2020

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. FACILITY NAME	2. DOCKET NUMBER	3. PAGE
Limerick Generating Station, Unit 2	05000353	1 OF 3

4. TITLE

Reactor Mode Switch Change During RCIC Testing

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
05	26	17	2017	- 007	- 00	11	28	17	FACILITY NAME	DOCKET NUMBER

9. OPERATING MODE: 1

10. POWER LEVEL: 008

11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: *(Check all that apply)*

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMBER *(Include Area Code)*
Robert B. Dickinson, Manager – Regulatory Assurance	(610) 718-3400

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

14. SUPPLEMENTAL REPORT EXPECTED	15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)* ☒ NO				

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On May 26, 2017, during the Limerick Unit 2 reactor startup from a refueling outage, the reactor mode switch was placed in the RUN position (OPCON 1) with Reactor Core Isolation Cooling (RCIC) inoperable for required surveillance testing due to the Control Room Supervisor (CRS) failing to maintain adequate control of plant evolutions prior to placing the Mode switch to RUN. The station initially determined this operation was permitted based on TS 3.0.4.c. On September 29, 2017, it was subsequently determined TS 3.0.4.c was inappropriately used for this evolution. Therefore, changing the Reactor Mode Switch from Startup to Run while RCIC was inoperable for testing was a TS violation and therefore reportable as an Operation or Condition Prohibited by TS under 10 CFR 50.73(a)(2)(i)(B). Operations will revise the 'Normal Plant Startup' procedure to incorporate lessons learned from this event. Additionally, a Shift Training Notification will be developed and distributed to ensure gaps and lessons learned are shared with the Operations organization.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Limerick Generating Station, Unit 2	05000353	2017	- 007	- 00

NARRATIVE

I. Unit Conditions Prior to the Event

Limerick Generating Station (LGS) Unit 2 was in Startup and being placed in the Run (Operational Condition (OPCON) 1) position at the time of the event. Reactor Core Isolation Cooling (RCIC) was inoperable for testing. There were no other structures, systems, or components inoperable at the time of the event that contributed to the event. There were no component failures as part of the event.

II. Description of the Event

During the Reactor startup from the 2017 Refueling Outage, 2R14, the Reactor Mode Switch was placed in the Run position while RCIC was inoperable for surveillance testing. The following is a timeline of events.

On May 26, 2017 at 16:10, Unit 2 RCIC was inoperable to perform the RCIC Comprehensive Surveillance Test, in accordance with Technical Specification (TS) 4.7.3.b.

On May 26, 2017 at 16:37, Unit 2 Reactor Mode Switch was placed in the RUN position.

Subsequently, the NRC questioned if TS allowed the station to change OPCONs while RCIC was inoperable. The station initially determined the condition was allowed based on TS 3.0.4.c, but that conclusion was later determined to be an inappropriate usage of the TS for the evolution.

On September 29, 2017, the Station determined that changing the Reactor Mode Switch from Startup to Run while RCIC was inoperable was a TS violation.

This evolution was determined to be an Operation or Condition Prohibited by TS and is reportable under 10 CFR 50.73(a)(2)(i)(B). This Licensee Event Report (LER) is being submitted within 60 days from the date of determination that this operation was prohibited by TS. Additionally, the NRC inspectors issued a Green Non-Cited Violation (NCV) as a result of this event. Specifically, a mode change from Startup to Run was made while TS 3.7.3 for RCIC was not met, and none of the allowances of TS 3.0.4 were met to allow the OPCON change. This NCV was issued as part of the third quarter integrated inspection, as documented in inspection report number 05000353/2017003.

III. Analysis of the Event

During the Unit 2 startup, RCIC was inoperable to perform the RCIC comprehensive test in accordance with TS 4.7.3.b. TS 4.7.3.b has an allowance that states the provisions of TS 4.0.4 are not applicable provided the surveillance is performed within 12 hours after reactor steam pressure is adequate to perform the test. TS 4.0.4 requires that the entry into an operating condition shall only be made when the TS LCO surveillance requirements have been met within their surveillance time interval. During the performance of the test, the plant OPCON was changed from Startup to Run.

The Station initially determined that the change was permitted by applying TS 3.0.4.c, since the TS LCO for RCIC was not met solely to perform the surveillance testing. Subsequent review concluded

NARRATIVE

that this determination was not correct. Specifically, TS 3.0.4 states that when an LCO is not met, entry into an operational condition shall only be made in accordance with TS 3.0.4.a, 3.0.4.b, or 3.0.4.c. However, the conditions for TS 3.0.4.a, 3.0.4.b, and 3.0.4.c are not applicable to TS 3.7.3.

IV. Safety Consequence

There was no actual safety consequence associated with this event. The RCIC system is a high-pressure coolant makeup system that is required for safe shutdown of the reactor whenever the reactor is isolated from its normal heat sink at elevated temperatures and pressures. TS Bases 3.7.3 identify that there is an increased risk associated with entering a mode with an inoperable RCIC subsystem. However, with the RCIC system inoperable, adequate core cooling is assured by the High Pressure Coolant Injection (HPCI) system, which was operable at the time, and therefore minimizes the safety significance of changing modes with RCIC inoperable.

V. Cause of the Event

The cause of the event was the Control Room Supervisor (CRS) failed to maintain adequate control of plant evolutions prior to placing the Mode switch to RUN.

Per TS Surveillance 4.7.3.b, the RCIC surveillance test must be completed within 12 hours of reaching adequate pressure. Based on past experiences, the Operations crew focused on ensuring the surveillance test was started as soon as possible after reaching adequate pressure to run the test. Concurrently, the plan for completing the final steps of the Operations Procedure for "Limerick Start Up Review" changed, causing a delay in changing OPCONs, as the RCIC test was initiated. The CRS did not adequately coordinate the two activities.

VI. Corrective Actions Completed/Planned

Performance management actions were applied for those personnel directly involved with the event. Operations will revise the 'Normal Plant Startup' procedure to incorporate lessons learned from this event to ensure effective coordination during future plant startups.

A Corrective Action Program (CAP) assignment was created to develop a Shift Training Notification to be distributed to ensure gaps and lessons learned are shared with the Operations organization.

VII. Previous Similar Occurrences

There have been no previous similar occurrences of changing OPCONs with RCIC or any Emergency Core Cooling System (ECCS) equipment inoperable at LGS in the last 5 years.